

Authority physical page 1 / folio 245 was read from the controlling raster. The running folio and copyright line are excluded from body prose; the journal, volume, page range, and year are retained as bibliographic metadata. The exact four-author byline, Griffiths’s star, the NSF note, and footnote 1 are preserved. No “Introduction” heading is inserted because none appears in the authority. Semantic text/LaTeX replay is sufficient; no crop is used.

French: “formal consequence” is rendered “conséquence formelle”; names, references, and mathematical types remain unchanged.

The displayed algebra maps have no printed labels: the left arrow is a plain left arrow and the right arrow is an inclusion hook. The lemma defines only $d^c = J^{-1}dJ$ on this page; no partial/bar-partial formula is imported. The prime in y' is preserved. Footnotes 2 and 3 remain attached to the Weil-conjecture and nilpotent-space statements. Semantic replay is sufficient; no crop is used.

French preserves the logical alternatives in hypothesis (2) and the inverse-system terminology.

The printed English reading “simplical” is retained diplomatically. Item (b) includes the factor $\otimes R$ after the parenthesized Q -polynomial forms. The introductory Laplacian identity on this page has exactly three terms, $\Delta_d = 2\Delta_\partial = 2\Delta_{\bar{\partial}}$; $\Delta_{\{d^c\}}$ is not added. Semantic replay is sufficient; no crop is used.

French uses the standard term “simplicial” while preserving all symbols and the scope of the tensor factor.

In the Section 7 preview, the authority reads E_1 and d_1 after stating degeneration at E_2 ; these indices are preserved. The source prints “principle fibrations”; that diplomatic reading is retained in English, while French uses the standard “fibrations principales”. In condition (1), the algebra A —not the differential d —is graded commutative. Semantic replay is sufficient; no crop is used.

The authority notation is $\Lambda(V_n)$, not $A(V_n)$, and the elementary extension uses $B = A \otimes \Lambda(V_k)$. The union is indexed $i \geq 0$. The degree-one condition is $M^1 = 0$, followed by the canonical sequence $M^{(2) \subset M}(3) \subset \cdots$. Footnote 4 is preserved. Semantic replay is sufficient; no crop is used. French uses “extension élémentaire”, “indécomposables”, and “série” consistently with the algebraic context.

The canonical series, dual Lie-algebra notation, and index placement were resolved against the raster: K , L_K , $[I(K)^i]^*$, L^i , and L_1^n are retained. The authority prints “a k -stage minimal model for M ” after $M_A \rightarrow A$; this apparent source anomaly is preserved rather than silently corrected. Semantic replay is sufficient; no crop is used.

French preserves the anomalous symbol M and records no conjectural emendation.

Authority physical page 7 / folio 251 was read from the controlling raster. Running head and folio are excluded. Preserve $\pi_*(\mathcal{A})$, not the inherited candidate's π_n . The sphere model's tensor has no printed subscript d , while the $\mathbb{C}P^n$ model does. Preserve $\mathcal{M}^{(k)} \subset \mathcal{M}^{(k+1)}$, $V_{k+1} = C_{k+1} \oplus N_{k+1}$, and the printed $\mathcal{M}^{(1)}$ series. Semantic replay is sufficient. French uses “algèbre 1-connexe”, “extension élémentaire”, and “noyau” consistently; formulas are unchanged. The authority writes the de Rham-complex subscripts S_n and $\mathbb{C}P_n$, while the cohomology groups use S^n and $\mathbb{C}P^n$; preserve this distinction.

Authority physical page 8 / folio 252 was read from the controlling raster. Preserve $\mathcal{E}_Y^* \xrightarrow{H^*} \mathcal{E}_X^* \otimes \mathcal{E}_I^*$, $\mathcal{N} = \mathcal{M} \otimes_d \Lambda(V_n)$, the obstruction σ , and the choice $\alpha(x)$; the inherited candidate's $e(x)$ is rejected. In Theorem 1.2 the solid arrows are $\mathcal{A} \xrightarrow[\textit{down}]{\varphi} \mathcal{B}$ and $\mathcal{M} \xrightarrow{f} \mathcal{B}$; inherited TeX reverses the labels. Corollary 1.5 prints $\rho_{\mathcal{B}} : \mathcal{M} \rightarrow \mathcal{B}$, although the diagram uses $\mathcal{M}_{\mathcal{B}}$; retain this source anomaly. The page ends at the diagram; do not add the inherited candidate's unprinted “commutes up to homotopy.”

Authority physical page 9 / folio 253 was read from the controlling raster. The three-line diagram is replayed semantically. The raw 1710×2580 authority image resolves the pasting operator as a tilde directly over an ordinary plus; encode $H \widetilde{+} H'$, not inherited $H \oplus -H'$. The codomain includes s, t, ds, dt . Preserve $\bar{u}, \bar{y} = \rho(y_{2n})$, and $d\bar{u} = f^*\bar{y}$; the inherited f_1/y rewrite is rejected. French preserves every formula and uses standard homotopy terminology.

Authority physical page 10 / folio 254 was read from the controlling raster. The source prints “simplical”; retain it diplomatically in English, while French uses standard “simplicial”. The authority actually prints $\tilde{f} : \mathcal{M}_N \cong \mathcal{M}_N$; context strongly suggests the second subscript should be M , but the source and French formulas retain the printed N rather than silently emending it. Preserve the lower-central-series placement Γ_2, Γ_1 and footnote 5.

Authority physical page 11 / folio 255 was read from the controlling raster. Preserve the switch from ψ_σ to ω_σ in the compatibility clause, the source readings “simplicial” and “We obtains”, and the absence of a summation sign in $\psi = \psi_{i_1\dots i_\ell}(t)dt_{i_1} \wedge \dots \wedge dt_{i_\ell}$. Stokes is exactly $\int_{\partial C} \psi = \int_C d\psi$. The theorem subtitle is read contextually as “p. l. de Rham Theorem”; the raster glyph itself is visually ambiguous between lowercase ell and numeral one. Preserve the diagram’s $\mathcal{E}_K^* \otimes_{\mathbb{Q}} \mathbb{R}$, $\mathcal{M}_{\mathcal{E}_X^*}^{(1)}$, and $\mathcal{L}_1^2(X) \rightarrow \mathcal{L}_1^1(X)$. French regularizes grammar only; formulas remain unchanged.

Authority physical page 12 / folio 256 was read from the controlling raster. In each parenthetical phrase and in the Section 3 bridge, the authority prints $\otimes \mathbb{R}$ without a subscript; the explicit algebraic formulas print $\otimes_{\mathbb{Q}} \mathbb{R}$. Preserve that distinction. The Postnikov diagram has vertical arrows p_2, p_1 , oblique arrows f_2, f_1 , and base arrow f_0 ; item 1 ends with a colon. French preserves all formulas and the exact logical scope.

Authority physical page 13 / folio 257 was read from the controlling raster; the running head and folio are excluded. Preserve the unaccented P_X , the printed wording “Towers of principle fibrations”, and the distinct tensor notations: the lemma has $\mathcal{A} \otimes_d H^*(F) \rightarrow \mathcal{E}_E^*$, the proof defines $\tilde{\rho} : \mathcal{A} \otimes H^*(F) \rightarrow \mathcal{E}_E^*$, and the comparison map is $\rho \otimes id : \mathcal{A} \otimes_d H^*(F) \rightarrow \mathcal{E}_B^* \otimes H^*(F)$. Transgression uses $\alpha \in H^k(F; \mathbb{Q})$ and $d\tilde{\alpha} = \pi^*(\beta)$; the principal-fibration example ends with $H^*(K(\pi, n); \mathbb{Q}) \cong \Lambda(\text{Hom}(\pi, \mathbb{Q}))$. The inherited TeX is retained as ZERO_ACCEPTED and is not promoted because it changes the transgressive datum and several displayed symbols.

Authority physical page 14 / folio 258 was read from the controlling raster; the running head and folio are excluded. Preserve $\Lambda_n = \Lambda(V_n^*)$ and $d : V_n^* \rightarrow \{\text{closed forms in } \mathcal{A}^{n+1}\} \rightarrow H^{n+1}(\mathcal{A})$. In $H^{n+1}(\mathcal{A}, V) \xrightarrow[\rho]{\cong} H^{n+1}(B; V)$, the congruence is above the arrow and ρ is beneath it. The characteristic-class diagram has $K(V, n) \rightarrow K(V, n)$ across the top, vertical arrows through E/P to $B/K(V, n + 1)$, $E \rightarrow P$ across the middle, and the bottom k -arrow; the three-stage composition diagram is also preserved. The inherited TeX remains ZERO_ACCEPTED and is not promoted because it substitutes $A(V_n^*)$ for $\Lambda(V_n^*)$ and corrupts the differential domain.

Authority physical page 15 / folio 259 was read from the controlling raster; the running head and folio are excluded. Preserve the reverse three-stage composition diagram and the long two-way correspondence $\mathcal{M}_X \longleftrightarrow P_X \otimes \mathbb{Q}$, not an isomorphism sign. The printed spelling “simplicial” is retained diplomatically. In the complex analogue, preserve the exact object $\pi_1(\mathcal{E}_M^* \otimes_{\mathbb{R}} \mathbb{C})$ and the associated minimal model $\mathcal{M}_{\mathcal{E}_M^* \otimes_{\mathbb{R}} \mathbb{C}}$. The inherited TeX remains ZERO_ACCEPTED because it changes the correspondence and substitutes a different fundamental-group object; its page-15 header-only detector crop is excluded from mathematical evidence.

Authority physical page 16 / folio 260 was read from the controlling raster; the running head and folio are excluded. Preserve the Massey representative $\eta \wedge \gamma + (-1)^{a+1} \alpha \wedge \rho$ in $H^{a+b+c-1} / \text{ideal}(a, c)$. Two visible grammatical anomalies are retained rather than silently regularized: definition (iii) says “a formal consequence of the cohomology M”, and the final definition says “a formal consequences of the induced map”. The homotopy-commutative square has $\mathcal{M}_{\mathcal{A}} \xrightarrow{\hat{f}} \mathcal{M}_{\mathcal{B}}$ above $H^*(\mathcal{A}) \xrightarrow{f^*} H^*(\mathcal{B})$, with vertical maps $\psi_{\mathcal{A}}$ and $\psi_{\mathcal{B}}$. The inherited TeX remains ZERO_ACCEPTED because it changes the Massey representative and regularizes printed wording.

Authority physical page 17 / folio 261 was read from the controlling raster; the running head and folio are excluded. Preserve the displayed upper-triangular matrix, $S^1 \rightarrow M^3 \rightarrow T^2$, and the printed model labels $\mathcal{M}_1^{(1)} = \Lambda_1(x, y)$ and $\mathcal{M}_2^{(1)} = \mathcal{M} = \Lambda_1(x, y) \otimes_d \Lambda_1(z)$, with $dz = x \wedge y$. In the 1-connected example, the generator order and differentials are $\Lambda_3(\psi, \varphi)$, $d\psi = x \wedge x$, and $d\varphi = x \wedge y$; the closed nonexact product is $x\varphi + y\psi$. Preserve $P[V_2 \oplus V_4 \oplus \cdots] \otimes \Lambda[V_1 \oplus V_3 \oplus V_5 \oplus \cdots]$ and Theorem 4.1 with $V_i = C_i \oplus N_i$ and $I(\oplus N_i)$. The inherited TeX remains ZERO_ACCEPTED because it changes the model labels, reverses the ψ/φ data, and omits terms from the algebra decomposition.

Authority physical page 18 / folio 262 was read from the controlling raster; the running head and folio are excluded. Preserve $J|T'(M) = \sqrt{-1}$ without an added identity operator, $T''(M)$ as the complex conjugate of $T'(M)$, and the exterior factor $\overline{\Lambda^q T'(M)^*}$. The form splitting is $\psi = \sum_{p,q} \psi_{p,q}$ and $\bigoplus_{p,q \geq 0} \mathcal{E}_M^{p,q}$. Preserve $d^c = \sqrt{-1}(\bar{\partial} - \partial)$, $(d^c)^2 = 0$, and $dd^c = 2\sqrt{-1} \partial \bar{\partial} = -d^c d$, followed by the unstarred cohomology notation $H_d(M)$ and $H_{d^c}(M)$. The inherited TeX remains ZERO_ACCEPTED because it rewrites the conjugate factor, changes ψ , omits $p, q \geq 0$, and alters the printed cohomology notation.

The controlling page topology is the polar-coordinate trio for d , d^c , and dd^c ; the Hermitian decomposition into S and A ; Laplacian identity (5.1); the metric and associated-form displays; and the two displayed parts of (5.2). Literal printed features include the pairing ending in $T(M)$, with no prime, and juxtaposition in the sum of ω_i with its conjugate. The inherited TeX omits the polar-coordinate trio, inserts a tensor product in that coframe sum, and collapses (5.2) into one numbered display; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the pair of defining Chern-connection equations; definitions (ii)-(iv); equation (5.3); the five-line partial-A computation; and the real-coordinate setup for (ii) implies (iii). The literal print has $\omega_i = \phi_i + \sqrt{-1} \phi_{(n+1)}$. The inherited TeX omits the separate nabla_H-double-prime equation, changes implication directions and notation, abbreviates the partial-A calculation, normalizes $n+1$ to $n+i$, and inserts a tensor product in the terminal metric sum; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the two displayed parts of (5.4); the 2-by-2 block connection matrix; the conjugate direct-sum identities for the complexified tangent bundle and connection; the biconditional (5.5); commutators (5.6) in their printed order; and the Laplacian expansion through (5.7). The inherited TeX merges (5.4), replaces the barred tangent summand by T-double-prime, weakens (5.5) to one direction, and reverses the order of the two zero commutators; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology runs from (5.8) through the four-term Laplacian calculation, identities (5.9)-(5.10), the three flat-coframe formulas, the cancellation proving (5.6), Lemma (5.11), and Hodge formulas (5.12)-(5.13). The printed lead-in for partial-star visibly ends with d^c lacking an adjoint star, while the following four-term expansion separately contains both $d^c d^c$ *and* $d^c d^c$. The inherited TeX suppresses that anomaly, substitutes A for Lambda in the Kahler identities, and changes the coframe and cancellation operators; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the Green-operator conclusion of the dd^c lemma, Remark (5.14), all six conditions in Lemma (5.15), and the componentwise implications at the page foot. Literal printed features include K^0 in the duality sentence and plain d in the application of $(a)_n$ to $d y^{(p-1,q)}$. The inherited TeX changes the inner Green-operator formula and replaces the printed K^0 duality indexing, although it retains the plain- d reading; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the final implication of Lemma (5.15); Remark (5.16) with five nodes and six arrows; Proposition (5.17), including the beta square of isomorphisms; the direct-sum filtration isomorphism; and decomposition formulas (1)-(3). Literal print has f_i on C_i to A , g_i on A to B_j , then $f_i g_j = 0$; repeats $K^{(p-1,q)}$ at the beta square's lower left; states $p+q-1=n$; opens the proof with numeric (1) implies (ii); prints $K^{(pq)}$; and retains $A^{(p,q)}$ in the d-prime parenthesis. The inherited TeX changes the map labels, six-arrow diagram, beta square, and filtration formula and normalizes $K^{(pq)}$; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the completion of Proposition (5.17); Remark (5.18); definitions (5.19) and (5.20), including their filtration, quotient, and sheaf-resolution displays; special case (5.21); and the opening of Theorem (5.22). Literal print calls the one-way implication in Remark (5.18) an equivalence, labels the first sheaf-resolution arrow with bar-partial, opens condition (5.21)(1) with the nested typography ((a), and retains $H^n(M;Q)$ for every n greater than or equal to zero. The inherited TeX strengthens the printed implication to a biconditional, omits the right arrow from $V^{p,q}$ to its ambient complexification, excludes weight zero by changing n greater than or equal to zero to n greater than zero, and changes the sheaf wedge to a tensor product; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the proof locator and idea for Theorem (5.22), Corollary (5.23), the two-clause Section 6 Main Theorem and both corollaries, then the first proof’s three complexes, its two-arrow i-and-rho diagram, the two claims, and proof steps (a) through (d). Literal print retains Mořsezon, the singular verb in ‘homotopy groups depends’, the equality order $d^c y=0=dy$, and the apparent sign sequence $dy=dd^c \omega$ followed by $x=y+d^c \omega$. The inherited TeX changes the printed d^c -Diagram Method heading to d -Diagram Method and renames the right-hand quotient map ρ as p ; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is proof step (e), its 2-by-3 functorial diagram and Q.E.D.; the second proof by the Principle of Two Types; the Massey-product pure-type decompositions and equations; the nested minimal-model sequence; and properties (1) through (5), which continue on the next page. Literal print uses the equivalence-to-zero sign in $\alpha \wedge \beta$ and gives no punctuation between the two formulas following ‘we may write in two ways’. The inherited TeX replaces that equivalence by equality, changes the minimal-model bound $i+j$ greater than or equal to $k+1$ to a strict inequality, and reverses all three downward f-star arrows in the functorial diagram; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the inductive extension by $V_{\{n+1\}}$; the kernel/cokernel split and Hodge-bigraded computations; the definitions of d , p , and p -prime on C and N ; the obstruction-killing argument; the final formality calculation; and the opening of Section 7, whose last sentence continues onto page 29. Literal print retains ‘non simply connected’ without a hyphen and ends the page mid-sentence after ‘spectral sequences’. The inherited TeX drops the equality case from the component sum $i+j$ greater than or equal to $r+1$ and replaces the final $F^{\{r-j+1\}}$ intersection $\bar{F}^j$ by $F^{\{i-j+1\}}$ intersection unbarred F^j ; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the carried-over word ‘collapse.’; the normal-crossing union and its differential algebra; the printed Meyer-Vietoris calculation; the tilde-D disjoint unions, alternating quotient, delta restriction, and double complex; the local exact sequence; the number-of-divisors spectral sequence with $|I|=q$ and d_1 from q to $q+1$; and Theorem (7.1) with its natural double-complex diagram. Literal print has ‘Meyer-Vietoris’, the isolated typo ‘double compexes’, and one $|I|=q$ abutment display. The inherited TeX changes α to ϕ , repeats D_i intersection D_j instead of printing D_j intersection D_i in the second restriction, removes all ten tilde-D marks, changes the delta sign exponent from ℓ to t , replaces the single $|I|=q$ abutment by two $|I|=q+1$ formulas, omits the double complex’s terminal arrow to zero, drops theorem number (7.1), and normalizes ‘compexes’; it remains ZERO_ACCEPTED and is not promoted.

The controlling page topology is the conclusion of Theorem (7.1), with E_k for 1 through infinity, vanishing differential, collapse at E_2 , and Q.E.D.; references 1 through 21; all four author address blocks; and the received date. Literal print retains ‘Bansfield’ in reference 1, ‘Moišezon’ in reference 8, ‘Isvestia’ in reference 21, and orders Whitehead as reference 16 before Whitney as reference 17. The inherited TeX adds a spurious prime to the second star of its E_1 and E_k bidegrees, changes Bansfield to Bousfield, swaps references 16 and 17, and omits all four address blocks and the received date; it remains ZERO_ACCEPTED and is not promoted.